

Module 8

End-to-End Project Workflow

Learning objectives

Bootstrap a project from a template repo

Run scripted build/test flows

Capture and analyze logs

Commit milestones and push for submission

Prerequisites

Modules 1–7 or equivalent experience

Learning path

Learning Path

Diagram: Learning Path
(render with mmdc when available)
flowchart LR
A[Clone template] --> B[Run scripts]
B --> C[Logs]
C --> D[Git commits]
D --> E[Submit]
E --> G[Done]

Course context

Full loop: edit → run → inspect → commit → push

Reproducibility matters for grading and debug

Pre-push checks catch common mistakes

Core topics

Self-check

```
$ ./scripts/module8.sh --check
```

Terminal: module_check

```

[0;36m=====
[0;36mModule 8: Environment self-check
[0;36m=====

[0;32m[OK] module8 directory exists: /home/yongfu/proj/universal-verification-methodology/learn
[0;32m[OK] template/sample_project exists
[0;32m[OK] sample_project/scripts/run_demo.sh exists
[0;32m[OK] logs/sample.log exists

[0;32m[OK] All required checks passed.
```

Run with log capture

```
./scripts/run_demo.sh > logs/run.log 2>&1  
grep -i error logs/run.log
```


Scripted workflow

```
$ cd module8/examples/template/sample_project && chmod u+x scripts/run_demo.sh && ./scripts/run_demo.sh
```

Terminal: sample_project_run

```
[2026-05-28T15:38:11+08:00] Run started
[INFO] Project root: /home/yongfu/proj/universal-verification-methodology/learn_unix_git/module8/exa
[INFO] Listing src/ and tb/
/home/yongfu/proj/universal-verification-methodology/learn_unix_git/module8/examples/template/sample
total 0
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 .
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 ..
-rwxrwxrwx 1 yongfu yongfu    0 May 28 12:03 .gitkeep

/home/yongfu/proj/universal-verification-methodology/learn_unix_git/module8/examples/template/sample
total 0
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 .
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 ..
-rwxrwxrwx 1 yongfu yongfu    0 May 28 12:03 .gitkeep
[2026-05-28T15:38:11+08:00] Run finished
```

Log inspection

```
$ grep INFO module8/examples/template/sample_project/logs/run.log 2>/dev/null | head -3
```

Terminal: grep_log

```
[INFO] Project root: /mnt/d/proj/universal-verification-methodology/learn_unix_git/module8/examples/  
[INFO] Listing src/ and tb/  
[INFO] Project root: /mnt/d/proj/universal-verification-methodology/learn_unix_git/module8/examples/
```

Practice

Exercises

Walk through sample_project/

Make a commit after a script run

Review pre-push checklist in module8

Summary & next steps

Unix + Git + scripts together

You are ready for design courses

Course complete — keep practicing

Next: Digital design & verification courses